How (95)–(97) are obtained

a step-by-step companion to Sec. VI-E

every step below is checked by `analysis/vi_e_geometric_box.py`

0. What is being bounded, and why the rate

The estimator does not see the tilt angle. It fits the two-segment model to the *rate*

$$\omega_{\text{nom}}(\tau) = C_1\big(\cosh C_2\tau - 1\big), \qquad C_1 = \frac{\dot M}{Wz_{CoM}}, \qquad C_2 = \sqrt{\frac{Wz_{CoM}}{J_P}}, \tag{D.1}$$

reconstructed from the gyro, and reports the onset as the argument that minimises the residual of that fit. Two questions follow, and (97) answers both:

(i) how far can the *true* rate depart from (D.1) before the fit stops describing the data? — the quantity $|e_\omega|_{\text{end}}/\omega_{\text{nom,end}}$;

(ii) how far does that departure move the *reported onset moment*? — the quantity $|\Delta M_{\text{crit}}|$.

These are different questions with different answers, which is why the two halves of (97) carry different constants ($\frac{1}{2}$ and 1 in the first, $\frac{1}{7}$ and $\frac{1}{5}$ in the second). Section 5 answers (i) and Section 6 answers (ii). If only one line of this document is read, let it be that one.

The departure obeys the Duhamel integral (88),

$$e_\omega(\tau) = \int_0^\tau \dot h_\varphi(\tau - s)\,\rho(s)\,ds, \qquad \dot h_\varphi(u) = \frac{\cosh C_2 u}{J_P}, \tag{D.2}$$

with ρ the part of the true forcing that the estimator's span $\{1, \tau, \delta\varphi\}$ cannot absorb — the gravity remainder ρ_φ and the bilinear ground-effect coupling ρ_{GE} of (89). Everything below is about turning (D.2) into a number.

1. The crude bound, and why it is not enough

Put $|\rho(s)| \le \rho_{\max}$ inside (D.2) and pull it out:

$$|e_\omega(\tau_{\text{end}})| \;\le\; \rho_{\max}\int_0^{\tau_{\text{end}}} \dot h_\varphi(\tau_{\text{end}} - s)\,ds \;=\; \rho_{\max}\int_0^{\tau_{\text{end}}} \frac{\cosh C_2 u}{J_P}\,du \;=\; \frac{\rho_{\max}\sinh x}{J_P C_2}, \quad x \triangleq C_2\tau_{\text{end}}. \tag{D.3}$$

The substitution $u = \tau_{\text{end}} - s$ was used in the middle step; the integral of cosh is sinh. Using $J_P = Wz_{CoM}/C_2^2$, i.e. $J_P C_2 = Wz_{CoM}/C_2$, the inertia disappears: $|e_\omega(\tau_{\text{end}})| \le \rho_{\max} C_2 \sinh x / Wz_{CoM}$. This is (90).

(D.3) is honest but blunt. It treats ρ as though it sat at its maximum for the whole window, whereas both channels start at zero at the onset and grow like τ^6 and τ^4. Evaluated over the geometric box of Sec. VI-D it returns a deviation of 446% at the slowest ramp — bigger than the signal it is supposed to bound. Sections 2–4 recover the lost factor.

2. Chebyshev's integral inequality

Statement. Let f, g be integrable on $[a, b]$. If they are *oppositely* monotone (one non-decreasing, the other non-increasing), then

$$\int_a^b fg \;\le\; \frac{1}{b-a}\int_a^b f \int_a^b g. \tag{D.4}$$

If they are monotone in the *same* direction, the inequality reverses. (Textbooks usually print the same-direction version, with $\ge$; the one needed here is the reversed one, so the direction is worth checking rather than remembering.)

Proof. Opposite monotonicity means $\big(f(u)-f(v)\big)\big(g(u)-g(v)\big) \le 0$ for every $u, v \in [a, b]$: when $u > v$ one bracket is ≥ 0 and the other ≤ 0. Integrate that product over the square $[a, b]^2$ and expand the four terms,

$$\underbrace{\iint f(u)g(u)}_{(b-a)\int fg} - \underbrace{\iint f(u)g(v)}_{\int f \int g} - \underbrace{\iint f(v)g(u)}_{\int f \int g} + \underbrace{\iint f(v)g(v)}_{(b-a)\int fg} \;=\; 2(b-a)\int fg - 2\int f \int g \;\le\; 0,$$

which is (D.4). $\square$

Application to (D.2). Take $[a, b] = [0, \tau_{\text{end}}]$ with $f(s) = \rho(s)$ and $g(s) = \dot{h}_\varphi(\tau_{\text{end}} - s)$.

- f is non-decreasing: both channels vanish at the onset and grow monotonically over the window ($\rho_\varphi \propto \varphi^2$ with φ rising, $\rho_{GE} \propto \tau\varphi$ likewise).
- g is non-increasing *in* s: as s grows the argument $\tau_{\text{end}} - s$ shrinks and cosh decreases. Note the kernel is increasing in its own argument; it is the reflection $s \mapsto \tau_{\text{end}} - s$ that makes it decreasing in s.

They are oppositely monotone, so (D.4) applies with $\le$, which is the useful direction:

$$e_\omega(\tau_{\text{end}}) \;\le\; \underbrace{\frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} \rho}_{\bar{\rho}} \cdot \int_0^{\tau_{\text{end}}} \dot{h}_\varphi(\tau_{\text{end}} - s)\, ds \;=\; \frac{\bar{\rho}\sinh x}{J_P C_2}. \tag{D.5}$$

(D.5) is (D.3) with $\rho_{\max}$ replaced by the *time average* $\bar{\rho}$. Nothing else changed. All that remains is to compute $\bar{\rho}/\rho_{\max}$ for the two channels.

3. The reduction factors R_φ and R_{GE}: (95)

Both channels have a known shape, because φ does. From (D.1), $\varphi_{\text{nom}}(\tau) = (C_1/C_2)\Lambda(C_2\tau)$ with $\Lambda(u) = \sinh u - u$. Hence, writing $u = C_2\tau$,

$$\rho_\varphi(\tau) = \tfrac{1}{2}G''\varphi^2 = A\,\Lambda(u)^2, \qquad \rho_{GE}(\tau) = \beta_M \dot{M}\tau\,\varphi = B\,u\,\Lambda(u),$$

with A, B constants whose values never matter — they cancel in the ratio below.

Gravity channel. The maximum is at the window end, $\rho_{\varphi,\max} = A\Lambda(x)^2$. The average is, substituting $u = C_2\tau$ so that $d\tau = du/C_2$ and $\tau_{\text{end}} = x/C_2$,

$$\bar{\rho}_\varphi = \frac{1}{\tau_{\text{end}}}\int_0^{\tau_{\text{end}}} A\Lambda(C_2\tau)^2 d\tau = \frac{C_2}{x}\cdot\frac{A}{C_2}\int_0^x \Lambda(u)^2 du = \frac{A}{x}\int_0^x \Lambda(u)^2 du,$$

so the constant A cancels and

$$R_\varphi(x) \;\triangleq\; \frac{\bar\rho_\varphi}{\rho_{\varphi,\max}} \;=\; \frac{1}{x}\,\frac{\int_0^x \Lambda(u)^2 du}{\Lambda(x)^2}. \tag{D.6}$$

Ground-effect channel. Identically, $\rho_{GE,\max} = B\, x\Lambda(x)$ and

$$R_{GE}(x) \;=\; \frac{1}{x}\,\frac{\int_0^x u\,\Lambda(u)\,du}{x\,\Lambda(x)}. \tag{D.7}$$

How to read $1/7$ and $1/5$. The quickest way to see where these come from — not the proof, which is below — is the order of each channel. Keeping the leading term of $\Lambda(u) = u^3/6 + u^5/120 + \cdots$ makes $\Lambda^2 \simeq u^6/36$ and $u\Lambda \simeq u^4/6$, so the two channels are sixth and fourth order near the onset, and a τ^n signal averages over $[0,T]$ to $T^n/(n{+}1)$ against its endpoint value T^n: hence $\frac{1}{7}$ and $\frac{1}{5}$. At the other end $\Lambda \to \sinh u \simeq e^u/2$ gives $\int_0^x \Lambda^2 \simeq e^{2x}/8$ and $\int_0^x u\Lambda \simeq xe^x/2$, so $R_\varphi \simeq 1/(2x)$ and $R_{GE} \simeq 1/x$ — the origin of the $\frac{1}{2}$ and 1 that Sec. 5 needs, once $\Psi \simeq x$ multiplies them.

Neither reading is used as a proof. (D.6b) below obtains the small-x limits from l'Hôpital's rule with no expansion of f whatever, and Sec. 5 obtains the large-x ones the same way. Only the monotonicity argument that follows uses a series, and only for its sign pattern.

Why the small-x value is the supremum. Both R's decrease throughout, and this needs no numerics. Substituting $u = xt$ in the numerator rescales the window to $[0,1]$:

$$R(x) = \frac{1}{x}\,\frac{\int_0^x f(u)\,du}{f(x)} = \int_0^1 \frac{f(xt)}{f(x)}\,dt. \tag{D.6a}$$

Let

$$n_{\rm eff}(u) \;\triangleq\; \frac{uf'(u)}{f(u)} \;=\; \frac{d\log f}{d\log u}$$

be the *local order* of f. The small-x limit of R follows from it by one application of l'Hôpital's rule, with no expansion of f: in the first form of (D.6a) numerator and denominator both vanish at $x = 0$, so

$$\lim_{x\to 0} R(x) = \lim_{x\to 0} \frac{\int_0^x f(u)\,du}{x\,f(x)} = \lim_{x\to 0} \frac{f(x)}{f(x) + xf'(x)} = \frac{1}{1 + n_{\rm eff}(0^+)}, \tag{D.6b}$$

the last step dividing through by f. Only the limit of the local order is needed, never the coefficients of f. Differentiating the integrand of (D.6a) logarithmically,

$$\frac{\partial}{\partial x}\log\frac{f(xt)}{f(x)} = t\,\frac{f'(xt)}{f(xt)} - \frac{f'(x)}{f(x)} = \frac{n_{\rm eff}(xt) - n_{\rm eff}(x)}{x}.$$

For $t \in (0,1)$ we have $xt < x$, so if $n_{\rm eff}$ is increasing the bracket is ≤ 0: every integrand of (D.6a) decreases in x, hence so does R. $\square$

Both channels satisfy the hypothesis, and with $\Lambda' = \cosh u - 1$ the local orders are expressible through the same $\Psi_\varphi(u) = u(\cosh u - 1)/(\sinh u - u) = u\Lambda'/\Lambda$ that Sec. 4 discarded as the wrong normaliser:

$$f = \Lambda^2: \quad n_{\rm eff} = 2\Psi_\varphi(u) \text{ from 6 to } \infty; \qquad f = u\Lambda: \quad n_{\rm eff} = 1 + \Psi_\varphi(u) \text{ from 4 to } \infty.$$

Why $n_{\rm eff}$ increases. What is required here is easy to understate, so it is worth naming precisely. Since $n_{\rm eff} = d\log f/d\log u$, the claim

$$\frac{dn_{\rm eff}}{d\log u} = \frac{d^2(\log f)}{d(\log u)^2} \;\ge\; 0$$

says that $\log f$ is *convex* in $\log u$ — that the slope of f on log–log axes grows. Monotonicity of f gives only $n_{\rm eff} \geq 0$, which is the slope's *sign* and nothing more, and the two properties are independent: $f(u) = 1 - e^{-u}$, $u/(1+u)$ and $\log(1+u)$ are all strictly increasing and all have $n_{\rm eff}$ strictly *decreasing*, hence R increasing. A saturating signal has a higher average relative to its endpoint the longer the window, not a lower one. The two channels here behave the opposite way not because they rise but because they *stiffen.*

Convexity does hold, and one place a power series is used — only for its sign pattern, with no coefficient evaluated — establishes it.

Lemma. Let $f(u) = \sum_{n\geq 0} a_n u^n$ converge for all $u \geq 0$ with every $a_n \geq 0$, and let f not be a single monomial. Then $n_{\rm eff} = uf'/f$ is strictly increasing on $(0, \infty)$.

Proof. Differentiating $n_{\rm eff}$,

$$u\,\frac{dn_{\rm eff}}{du} = \frac{uf'}{f} + \frac{u^2 f''}{f} - \Big(\frac{uf'}{f}\Big)^2 = \frac{S_0 S_2 - S_1^2}{S_0^2}, \qquad S_k \triangleq \sum_n n^k a_n u^n,$$

since $uf' = S_1$ and $u^2 f'' = S_2 - S_1$, so the first two terms combine to S_2/S_0. Cauchy–Schwarz applied to the two sequences $\{\sqrt{a_n u^n}\}$ and $\{n\sqrt{a_n u^n}\}$ — legitimate because $a_n u^n \geq 0$, which is the only place the sign pattern is needed — gives $S_1^2 \leq S_0 S_2$. Hence $dn_{\rm eff}/du \geq 0$, with equality only when the two sequences are proportional, i.e. only when a single a_n is non-zero. □

(Dividing through by S_0 turns S_k/S_0 into the moments of the weights $a_n u^n/f$ and the numerator into their variance, which is a shorter way to say the same thing; nothing below depends on reading it that way.)

Both kernels qualify: $\Lambda = \sum_{k\geq 1} u^{2k+1}/(2k+1)!$ has non-negative coefficients, and so do the product Λ^2 and the shift $u\Lambda$, and neither is a monomial. So $n_{\rm eff}$ is strictly increasing for both channels and R is strictly decreasing. The same weights give the origin values for free, since as $u \to 0$ they concentrate on the lowest exponent present, 6 and 4.

The same fact as a closed-form derivative. An appendix that quotes dR/dx explicitly needs those expressions signed, and the argument above does it with no further machinery — in particular without the twelve-stage differentiation that signing them directly would cost. Write $A(x) = \int_0^x f$, so $R = A/(xf)$. Then, using $A' = f$ and $xf' = n_{\rm eff} f$,

$$R'(x) = \frac{A'\,xf - A\,(f + xf')}{x^2 f^2} = \frac{xf(x) - \big(1 + n_{\rm eff}(x)\big)A(x)}{x^2 f(x)}. \tag{D.7a}$$

The sign turns on whether A exceeds $xf/(1 + n_{\rm eff})$, and it does, for exactly the reason $n_{\rm eff}$ increases. Since $d\log f/d\log u = n_{\rm eff}$ and $n_{\rm eff}(t) \leq n_{\rm eff}(x)$ for $t \leq x$,

$$\log\frac{f(x)}{f(u)} = \int_u^x n_{\rm eff}(t)\,\frac{dt}{t} \;\leq\; n_{\rm eff}(x)\,\log\frac{x}{u} \quad\Longrightarrow\quad f(u) \;\geq\; f(x)\Big(\frac{u}{x}\Big)^{n_{\rm eff}(x)},$$

and integrating that minorant over $[0, x]$,

$$A(x) \;\geq\; f(x)\int_0^x \Big(\frac{u}{x}\Big)^{n_{\rm eff}(x)} du \;=\; \frac{x\,f(x)}{1 + n_{\rm eff}(x)},$$

strictly, since $n_{\rm eff}$ is not constant. The numerator of (D.7a) is therefore negative and $R' < 0$. □

In words: f is at least as steep as the power law of its local order at x, so its average over the window is at least the $1/(n{+}1)$ that power law would give — and $1/(n{+}1)$ is precisely the value R would need for R' to vanish.

Substituting $n_{\rm eff} = 2x\Lambda'/\Lambda$ for $f = \Lambda^2$, and $n_{\rm eff} = (\Lambda + x\Lambda')/\Lambda$ for $f = u\Lambda$, and clearing denominators puts (D.7a) in the form an appendix would quote:

$$\frac{dR_\varphi}{dx} = \frac{x\Lambda^3 - (\Lambda + 2x\Lambda')\int_0^x \Lambda^2 du}{x^2\Lambda^3}, \qquad \frac{dR_{GE}}{dx} = \frac{x^2\Lambda^2 - (2\Lambda + x\Lambda')\int_0^x u\Lambda\, du}{x^3\Lambda^2}, \tag{D.7b}$$

both negative for $x > 0$. The factors of x inside the brackets are load-bearing: with them the numerators cancel to order x^{12} and x^{10} at the origin, which is what leaves R' finite there; drop either and the expression diverges as x^{-2} instead. Both forms are checked against a five-point difference of R by `analysis/closed_form_check.py`.

The variance route needs the series; the following one does not, and is worth recording because it settles the same two numbers independently. The value of Ψ_φ at the origin is l'Hôpital's rule, applied three times. Writing $\Psi_\varphi = N/D$ with $N = u\cosh u - u$ and $D = \sinh u - u$, successive differentiation gives the four pairs

$$\frac{u\cosh u - u}{\sinh u - u}, \qquad \frac{\cosh u + u\sinh u - 1}{\cosh u - 1}, \qquad \frac{2\sinh u + u\cosh u}{\sinh u}, \qquad \frac{3\cosh u + u\sinh u}{\cosh u},$$

of which the first three are $0/0$ at $u = 0$ while the fourth is $3/1$. Hence $\Psi_\varphi(0^+) = 3$, so $n_{\text{eff}}(0^+)$ is 6 for Λ^2 and 4 for $u\Lambda$, and (D.6b) returns $\frac{1}{7}$ and $\frac{1}{5}$. The mechanism is physical: Λ behaves as u^3 near the onset and grows exponentially later, so the larger the window the more sharply both channels are concentrated at its end, and the smaller their average is relative to the endpoint value. The small-x values are therefore suprema:

$$\bar\rho = R_\varphi(x)\,\rho_{\varphi,\max} + R_{GE}(x)\,\rho_{GE,\max}, \qquad R_\varphi \le \tfrac{1}{7}, \qquad R_{GE} \le \tfrac{1}{5}. \tag{95}$$

The suprema are approached but never attained: for $x > 0$ the inequalities are strict. Over the flown $x \in [1.79, 5.20]$ the factors run 0.089–0.133 and 0.145–0.191, so quoting $\frac{1}{7}$ and $\frac{1}{5}$ gives away 10–60% — while gaining a factor of 5 to 11 over (D.3).

4. Normalising on the rate: (96)

Divide (D.5) by the nominal rate at the same instant, $\omega_{\text{nom,end}} = C_1(\cosh x - 1)$ with $C_1 = \dot M / W z_{CoM}$, and use $J_P C_2 = W z_{CoM}/C_2$:

$$\frac{|e_\omega|_{\text{end}}}{\omega_{\text{nom,end}}} \le \frac{\bar\rho\, C_2 \sinh x / W z_{CoM}}{(\dot M / W z_{CoM})(\cosh x - 1)} = \frac{\bar\rho\, C_2}{\dot M} \cdot \frac{\sinh x}{\cosh x - 1}.$$

$W z_{CoM}$ has cancelled. Now eliminate $\dot M$ in favour of the moment increment the window actually spans. Since $\tau_{\text{end}} = x/C_2$,

$$\Delta M_{\text{win}} = \dot M \tau_{\text{end}} = \frac{\dot M x}{C_2} \qquad \Longleftrightarrow \qquad \frac{C_2}{\dot M} = \frac{x}{\Delta M_{\text{win}}},$$

and substituting,

$$\frac{|e_\omega|_{\text{end}}}{\omega_{\text{nom,end}}} \le \frac{\Psi(x)\,\bar\rho}{\Delta M_{\text{win}}}, \qquad \Psi(x) = \frac{x\sinh x}{\cosh x - 1}. \tag{96}$$

The half-angle identities $\cosh x - 1 = 2\sinh^2(x/2)$ and $\sinh x = 2\sinh(x/2)\cosh(x/2)$ collapse the quotient:

$$\Psi(x) = x\,\frac{2\sinh(x/2)\cosh(x/2)}{2\sinh^2(x/2)} = x\coth\frac{x}{2} \;\ge\; 2.$$

Ψ rises from 2 at $x \to 0$ to x as $x \to \infty$. (Had the excursion been used as the normaliser instead of the rate, the factor would have been $x(\cosh x - 1)/(\sinh x - x)$, rising from 3 — a different function, and the wrong one, since the estimator never looks at φ.)

5. Why $\Psi\bar{\rho}$ stays bounded: first half of (97)

Ψ grows with x and the R's shrink with it, and the two products converge to $\frac{1}{2}$ and 1. Both bounds are exact, and neither needs the series: two differentiations settle each.

The claims in integral form. Since $\Psi = x\coth(x/2)$ and $\rho_{\varphi,\max} = A\Lambda(x)^2$,

$$\Psi R_\varphi = x\coth\tfrac{x}{2}\cdot\frac{\int_0^x \Lambda^2 du}{x\Lambda(x)^2} = \coth\tfrac{x}{2}\,\frac{\int_0^x \Lambda^2 du}{\Lambda(x)^2},$$

so $\Psi R_\varphi \le \frac{1}{2}$ is exactly

$$\int_0^x \Lambda(u)^2 du \ \le\ \tfrac{1}{2}\tanh\tfrac{x}{2}\,\Lambda(x)^2, \tag{D.10}$$

and the same reduction turns $\Psi R_{GE} \le 1$ into

$$\int_0^x u\,\Lambda(u)\,du \ \le\ x\tanh\tfrac{x}{2}\,\Lambda(x). \tag{D.11}$$

The half-angle tangent is not an accident: $\tanh(x/2) = (\cosh x - 1)/\sinh x = \Lambda'(x)/\sinh x$, so both statements compare an integral of Λ against Λ and its own derivative.

Where $\frac{1}{2}$ and 1 come from. Both are the $x\to\infty$ limits, and each is one application of l'Hôpital's rule. Written as above, both products have the form $\coth(x/2)\int_0^x f/f(x)$ — with $f = \Lambda^2$ for the gravity channel and $f = u\Lambda$ for the ground-effect channel — and $\coth(x/2)\to 1$. In the remaining ratio numerator and denominator both diverge, so

$$\lim_{x\to\infty}\frac{\int_0^x f(u)\,du}{f(x)} = \lim_{x\to\infty}\frac{f(x)}{f'(x)} = \Big(\lim_{x\to\infty}\frac{f'}{f}\Big)^{-1},$$

and the two logarithmic derivatives are elementary:

$$\frac{f'}{f}\bigg|_{f=\Lambda^2} = \frac{2\Lambda'}{\Lambda} = \frac{2(\cosh x - 1)}{\sinh x - x} \ \longrightarrow\ 2, \qquad \frac{f'}{f}\bigg|_{f=u\Lambda} = \frac{1}{u} + \frac{\Lambda'}{\Lambda} \ \longrightarrow\ 1,$$

giving $\frac{1}{2}$ and 1. Each constant is thus the reciprocal of its kernel's exponential growth rate, which is what $\int_0^x e^{ku}du = e^{kx}/k$ says: the gravity channel gets the smaller constant because its kernel, $\Lambda^2 \sim e^{2x}/4$, is the square of the other's, $u\Lambda \sim u\,e^x/2$.

The only theorem used. Everything below rests on one corollary of the mean value theorem, applied three times and nowhere supplemented.

Lemma. If f is continuous on $[0,b]$, differentiable on $(0,b)$ with $f'\ge 0$ there, and $f(0)=0$, then $f\ge 0$ on $[0,b]$. *Proof:* for $0\le s<t\le b$ the mean value theorem gives $f(t)-f(s) = f'(\xi)(t-s)\ge 0$, so f is non-decreasing; taking $s=0$ gives $f(t)\ge f(0)=0$. $\square$

Note what is *not* used. No limit as $x\to\infty$ enters: the constant $\frac{1}{2}$ comes from the algebra of (D.10), and the inequality is established at each x separately. No monotonicity of ΨR_φ is assumed or proved. The limits quoted above serve only to show the constants cannot be improved.

Proof of (D.10). Let $D(x)$ be the right side minus the left, so that (D.10) is $D\ge 0$. Every quantity divided by below — $\coth(x/2)$, Λ^2, Λ, $\cosh^2(x/2)$ — is strictly positive for $x>0$, so each step is an equivalence.

$D(0)=0$, since $\tanh 0 = \Lambda(0) = \int_0^0 = 0$. Differentiating and factoring out $\Lambda>0$,

$$D'(x) = \Lambda(x)\,B(x), \qquad B(x) = \tfrac{1}{4}\operatorname{sech}^2\tfrac{x}{2}\,\Lambda + \tanh\tfrac{x}{2}\,\Lambda' - \Lambda,$$

so $D'\ge 0$ iff $B\ge 0$. Write $y = x/2$, $S=\sinh y$, $C=\cosh y$, giving $\Lambda = 2SC - 2y$ and $\Lambda' = 2S^2$. Multiplying B by $2C^2>0$,

$$2C^2 B = (SC - y) + 4S^3C - 4SC^3 + 4yC^2,$$

and since $S^3C - SC^3 = SC(S^2 - C^2) = -SC$ the two cubic terms collapse to $-4SC$, leaving $y(4C^2 - 1) - 3SC = y(3 + 4S^2) - 3SC$. Now $4S^2 = 2\cosh x - 2$ and $2SC = \sinh x$ and $y = x/2$, so

$$B(x) = \frac{\phi(x)}{4\cosh^2(x/2)}, \qquad \phi(x) \triangleq x + 2x\cosh x - 3\sinh x,$$

and $B \geq 0$ iff $\phi \geq 0$. Finally

$$\phi'(x) = 1 - \cosh x + 2x\sinh x, \qquad \phi''(x) = \sinh x + 2x\cosh x.$$

First application. $f = \phi'$: both terms of $\phi'' = \sinh x + 2x\cosh x$ are non-negative for $x \geq 0$, and $\phi'(0) = 1 - 1 + 0 = 0$. The Lemma gives $\phi' \geq 0$.

Second application. $f = \phi$: $\phi' \geq 0$ by the first application and $\phi(0) = 0$. The Lemma gives $\phi \geq 0$, hence $B \geq 0$, hence $D' = \Lambda B \geq 0$.

Third application. $f = D$: $D' \geq 0$ by the second application and $D(0) = 0$. The Lemma gives $D \geq 0$, which is (D.10). □

Proof of (D.11). The same Lemma settles the ground-effect channel, and here it is needed only once. Let E be the right side of (D.11) minus the left,

$$E(x) \;=\; x\tanh\tfrac{x}{2}\,\Lambda(x) \;-\; \int_0^x u\,\Lambda(u)\,du, \qquad E(0) = 0.$$

Differentiating — the integral contributes $x\Lambda(x)$ — and using $\operatorname{sech}^2(x/2) = 2/(\cosh x + 1)$ and $\tanh(x/2) = \sinh x/(\cosh x + 1)$ to put everything over $\cosh x + 1$, the hyperbolic terms collapse to

$$E'(x) \;=\; \frac{\sinh^2 x - 2x\sinh x + x^2\cosh x}{\cosh x + 1} \;=\; \frac{\Lambda(x)^2 + x^2\Lambda'(x)}{\cosh x + 1}, \tag{D.12}$$

the second equality being the identity $(\sinh x - x)^2 + x^2(\cosh x - 1) = \sinh^2 x - 2x\sinh x + x^2\cosh x$. The numerator is a sum of two non-negative terms and the denominator is positive, so $E' \geq 0$; the Lemma gives $E \geq 0$, which is (D.11). □

No series is needed for either, and (D.11) costs one application of the Lemma against three for (D.10) — the completed square in (D.12) does the work that ϕ did there.

No monotonicity of the products is claimed or needed — only that they do not exceed their limits, which is what (D.10) and (D.11) say. Expanding $\bar{\rho}$ by (95),

$$\Psi\bar{\rho} = \big(\Psi R_\varphi\big)\rho_{\varphi,\max} + \big(\Psi R_{GE}\big)\rho_{GE,\max} \;\leq\; \tfrac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max},$$

which is the left half of (97). *x has disappeared* — and with it C_2, J_P and Wz_{CoM}. This is the whole point of the exercise: the bound no longer depends on the window length, so it does not have to be solved for, only bounded well enough to keep $\rho_{GE,\max}$ finite (which is what (92)–(93) do).

6. From the rate deviation to the threshold: second half of (97)

The reported quantity is not e_ω but the onset moment, so the deviation must be pushed one step further.

How the onset responds to a perturbation. The estimator picks t_0 to minimise $J(t_0) = \int\big(y(t) - \hat{\omega}(t;t_0)\big)^2 dt$ with $\hat{\omega}(t;t_0) = C_1(\cosh C_2(t - t_0) - 1)$. Write the data as the nominal solution plus the deviation, $y = \hat{\omega}(t;t_0^*) + e_\omega$, and the minimiser as $t_0 = t_0^* + \delta$. To first order $\hat{\omega}(t;t_0^* + \delta) \simeq \hat{\omega}(t;t_0^*) + \delta\,\partial_{t_0}\hat{\omega}$, so the residual is $e_\omega - \delta\,\partial_{t_0}\hat{\omega}$ and stationarity $\partial J/\partial t_0 = 0$ reads

$$\int\big(e_\omega - \delta\,\partial_{t_0}\hat{\omega}\big)\partial_{t_0}\hat{\omega}\,dt = 0 \qquad\Longrightarrow\qquad \delta = \frac{\langle e_\omega,\ \partial_{t_0}\hat{\omega}\rangle}{\|\partial_{t_0}\hat{\omega}\|^2}.$$

With $\partial_{t_0}\hat{\omega} = -C_1 C_2 \sinh(C_2\tau)$ and $\Delta M_{\text{crit}} = \dot{M}\delta$, and using $C_1 = \dot{M}/Wz_{CoM}$,

$$\Delta M_{\text{crit}} = -\frac{Wz_{CoM}\,\langle e_\omega,\ \sinh(C_2\tau)\rangle}{C_2\,\|\sinh(C_2\tau)\|^2}. \tag{D.8}$$

This is exactly the reduction `error_budget.py` implements.

It is an average of ρ with a non-negative weight. Substitute the Duhamel integral (D.2) into (D.8) and exchange the order of integration (Fubini; both integrands are continuous on a bounded domain):

$$\int_0^T \sinh(C_2\tau)\int_0^\tau \dot{h}_\varphi(\tau-s)\rho(s)\,ds\,d\tau = \int_0^T \rho(s)\,\underbrace{\left[\int_s^T \sinh(C_2\tau)\,\dot{h}_\varphi(\tau-s)\,d\tau\right]}_{\text{depends on } s \text{ only}} ds,$$

the inner limits changing from $s \in [0,\tau]$ to $\tau \in [s,T]$. Hence

$$\Delta M_{\text{crit}} = -\int_0^T \rho(s)\,w(s)\,ds, \qquad w(s) = \frac{Wz_{CoM}\displaystyle\int_s^T \sinh(C_2\tau)\cosh\big(C_2(\tau-s)\big)d\tau}{J_P C_2\|\sinh(C_2\tau)\|^2}. \tag{D.9}$$

Three properties of w finish the argument, and all three are checked in code (`error_budget.py --check`):

1. $w \geq 0$, because the integrand is a product of two hyperbolic functions of non-negative arguments;

2. $\int_0^T w = 1$, exactly. This is not a numerical coincidence and is worth doing, because it is what lets $\bar{\rho}$ enter (97) bare. Integrate the numerator of (D.9) over s and exchange the order: the region $0 \leq s \leq \tau \leq T$ is the same triangle read either way, so

$$\int_0^T\int_s^T \sinh(C_2\tau)\cosh\big(C_2(\tau-s)\big)d\tau\,ds = \int_0^T \sinh(C_2\tau)\left[\int_0^\tau \cosh\big(C_2(\tau-s)\big)ds\right]d\tau.$$

   The inner integral is elementary — substitute $u = \tau - s$, whose sign flip is cancelled by the reversed limits —

$$\int_0^\tau \cosh\big(C_2(\tau-s)\big)ds = \int_0^\tau \cosh(C_2 u)\,du = \frac{\sinh(C_2\tau)}{C_2},$$

   and putting it back reproduces the very norm that sits in the denominator of (D.9),

$$\frac{1}{C_2}\int_0^T \sinh^2(C_2\tau)\,d\tau = \frac{\|\sinh(C_2\tau)\|^2}{C_2}.$$

   Hence

$$\int_0^T w = \frac{Wz_{CoM}\,\|\sinh\|^2/C_2}{J_P C_2\|\sinh\|^2} = \frac{Wz_{CoM}}{J_P C_2^2} = 1$$

   by $J_P C_2^2 = Wz_{CoM}$ — the same identity that removes the inertia everywhere else in this document. Neither the window length nor any constant survives. `error_budget.py --check` evaluates (D.9) at $\rho \equiv 1$ and returns -1.0005 to -1.0001 across five operating points, which is the quadrature error on an identity rather than evidence for it. The reading is that a uniform unmodelled moment displaces the identified threshold one-for-one, and now that is a consequence rather than an assumption;

3. w is non-increasing in s — as s grows both the integration range and the cosh factor shrink (verified numerically over the window).

Property 1 plus 2 alone would give $|\Delta M_{\text{crit}}| \le \rho_{\max}$. But ρ is increasing and w is decreasing, so *Chebyshev applies a second time*, now to (D.9):

$$\Big|\int_0^T \rho\, w\Big| \;\le\; \frac{1}{T}\int_0^T \rho \int_0^T w \;=\; \bar\rho\cdot 1 \;=\; \bar\rho,$$

and substituting (95),

$$|\Delta M_{\text{crit}}| \;\le\; \bar\rho \;\le\; \frac{\rho_{\varphi,\max}}{7} + \frac{\rho_{GE,\max}}{5}, \tag{97b}$$

the right half of (97).

Why the constants differ from the left half. The left half asks how far the rate curve is displaced, so the kernel integral Ψ multiplies $\bar\rho$ and the products ΨR climb to $\frac{1}{2}$ and 1. The right half asks how far the *minimiser* moves, and the minimiser is found by correlating against $\sinh(C_2\tau)$ — a normalised operation, $\int w = 1$, with no kernel gain left over. So there $\bar\rho$ enters bare, and the constants are the R's themselves, $\frac{1}{7}$ and $\frac{1}{5}$. The threshold bound is the stronger of the two by roughly a factor of four, which is why it, and not the relative rate bound, is the number carried into the offset budget.

7. A worked example, checkable by hand

Take the roll axis at the fastest ramp, evaluated over the geometric box of Sec. VI-D rather than at any fitted constant: $W = 31.59$ N (fully ballasted, $m = 3.221$ kg), $z_{CoM} = 0.30$ m, $a = l_p + p_{\text{off}} = 0.160$ m, $\dot M = 1.20$ N m/s, $\varphi_{\max} = 10° = 0.17453$ rad, $\beta_M = 0.0345$ rad^{-1}, $J_{CoM}^{\text{CAD}} = 0.0537$ kg m^2.

step	expression	value
geometry	$Wz_{CoM} = W z_{CoM}$	9.477 N m
(82) lower	$J_P \ge m(z_{CoM}^2 + a^2)$	0.3724 kg m^2
(82) upper	$J_P \le J_{CoM}^{\text{CAD}} + m(z_{CoM}^2 + a^2)$	0.4261 kg m^2
	$C_2 \le \sqrt{g z_{CoM}/(z_{CoM}^2 + a^2)}$	5.045 s^{-1}
tilt cap → Λ	$\Lambda_{\max} = \varphi_{\max} W z_{CoM} C_2/\dot M$	6.954
(92)	$\bar x = (6\Lambda_{\max})^{1/3}$	3.468
exact	solve $\sinh x - x = \Lambda_{\max}$	$x = 2.993$
(93)	$\tau_{\text{end}} \le (6\varphi_{\max} J_P/\dot M)^{1/3}$	0.719 s
(93)	$\Delta M_{\text{win}} = \dot M \tau_{\text{end}}$	0.863 N m
channel	$\rho_{\varphi,\max} = \frac{1}{2} W a\, \varphi_{\max}^2$	76.98 mN m
(93)	$\rho_{GE,\max} = \lvert\beta_M\rvert \Delta M_{\text{win}} \varphi_{\max}$	5.20 mN m
(D.6)	$R_\varphi(3.468)$	0.1125
(D.7)	$R_{GE}(3.468)$	0.1705
(95)	$\bar\rho = R_\varphi \rho_{\varphi,\max} + R_{GE}\rho_{GE,\max}$	9.54 mN m
	(against $\rho_{\max} = 82.2$ mN m)	
(96)	$\Psi(\bar x) = \bar x \coth(\bar x/2)$	3.692
(96)	$\Psi\bar\rho/\Delta M_{\text{win}}$	4.08%
(97)	$(\frac{1}{2}\rho_{\varphi,\max} + \rho_{GE,\max})/\Delta M_{\text{win}}$	5.06%
(97b)	$\rho_{\varphi,\max}/7 + \rho_{GE,\max}/5$	12.04 mN m
	$\div W_{\min}$	0.400 mm
(D.3)	envelope only, $\Psi\rho_{\max}/\Delta M_{\text{win}}$	35.2%

Two readings. First, (97) costs about a quarter over (96) — the price of dropping x from the statement — whereas the envelope form costs a factor of nine. Second, this is the *worst admissible* case: the box is evaluated at the 10° tilt cap, at the top of the z_{CoM} range, and with the inertia at whichever end of (82) is unfavourable for the quantity being bounded. Over the runs actually flown the excursions are 5–9° and the exact per-run propagation, which convolves the measured ρ instead of any envelope, gives 0.04 mN m in the median and 0.15 at worst — 0.001 to 0.004 mm of offset.

8. Summary of the chain

	step	what it buys
(D.2)	$e_\omega = \int \dot{h}_\varphi(\tau_{\text{end}} - s)\rho(s)ds$	the exact deviation
(D.3)	$\lvert\rho\rvert \le \rho_{\max}$, integrate the kernel	(90); too weak by $\sim 8\times$
(D.4)	Chebyshev, $\rho\uparrow$ against $\dot{h}_\varphi(\tau_{\text{end}} - \cdot)\downarrow$	$\rho_{\max} \to \bar{\rho}$
(D.6–7)	known shapes Λ^2 and $u\Lambda$	the ratios $\bar{\rho}/\rho_{\max}$
(D.6a)	$R = \int_0^1 f(xt)/f(x)dt$, $n_{\text{eff}}\uparrow$	(95): $R_\varphi \le \frac{1}{7}$, $R_{GE} \le \frac{1}{5}$
(96)	divide by $\omega_{\text{nom,end}}$, use $\Delta M_{\text{win}} = \dot{M}x/C_2$	$\Psi = x\coth(x/2)$; Wz_{CoM}, J_P gone
	$\Psi R_\varphi \uparrow \frac{1}{2}$, $\Psi R_{GE} \uparrow 1$	(97) left half; x gone
(D.8–9)	linearise the onset, Fubini, $w \ge 0$, $\int w = 1$	$\Delta M_{\text{crit}} = -\langle\rho\rangle_w$
	Chebyshev again, $\rho\uparrow$ against $w\downarrow$	(97) right half: $\frac{1}{7}$, $\frac{1}{5}$

Chebyshev is used twice, for the same reason both times: the forcing is concentrated at the end of the window and every kernel in the problem weights the end of the window least. The two uses are independent — the first converts a kernel integral, the second a normalised correlation — which is why the constants they leave behind differ.